Analysis of Air Pollution Over a Year

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DS 4002: March 23, 2026

Outline

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Project Details

- **Motivation:** Air quality is a critical public health concern, as rising pollutant levels impact human health and climate change. This has resulted in monitoring stations that collect data on compound concentrations, temperature, and humidity across cities worldwide, making time series analysis essential for identifying air quality and pollution trends.
- **Hypothesis:** All pollution levels will be higher during the day than during the night and during the week compared to the weekend.
- **Research Questions:** Is pollution higher during the day than by night? Is pollution higher during weekdays than weekends? Which pollutions levels correlate most with these cycles?
- **Modeling Approach:** Conduct statistical and linear regression analysis on pollution levels, along with applying the Pearson correlation coefficient for statistical significance.
- **Goal:** Investigate whether the air quality differs based on the hour of day and type of day from the UC Irvine dataset

Data Acquisition & Explanation

Acquisition: UC Irvine Archives

Summary of Established Data Set:

- 9357 rows, 15 columns
- Appended both ***"Day"*** and ***"Weekday"*** boolean columns to indicate day vs. night and type of day

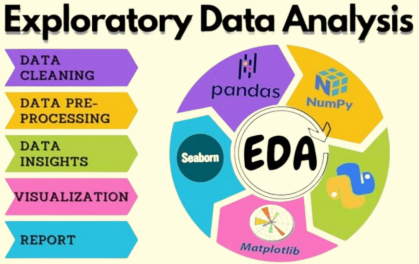
Column Name	Description	Potential Response
Date	Date of recording	10/03/2004
Time	Time of recording	18.00.00
Day	Daytime flag: 1 for day, 0 for night	1, 0
Weekday	Weekday flag: 1 for weekday, 0 for weekend	1, 0
CO(GT)	True hourly averaged CO concentration in mg/m^3 (reference analyzer)	2.6, 2.0, -200 (missing)
PT08.S1(CO)	Hourly averaged tin oxide sensor response, nominally CO targeted	1360, 1292
NMHC(GT)	True hourly averaged Non-Methanic Hydrocarbons concentration in $\mu g/m^3$ (reference analyzer)	150, 112, -200 (missing)
C6H6(GT)	True hourly averaged Benzene concentration in $\mu g/m^3$ (reference analyzer)	11.9, 9.4
PT08.S2(NMHC)	Hourly averaged titania sensor response, nominally NMHC targeted	1046, 955

... more compounds

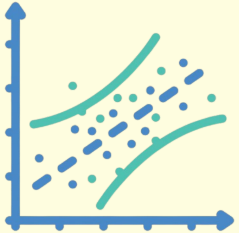
Analysis Plan & Justification

Retrieve data from **UC Irvine repository**

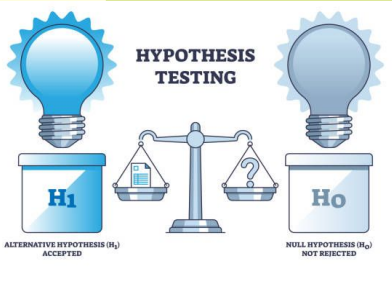
Replace -200 code with NaN



Perform **EDA** to get a sense of data distribution

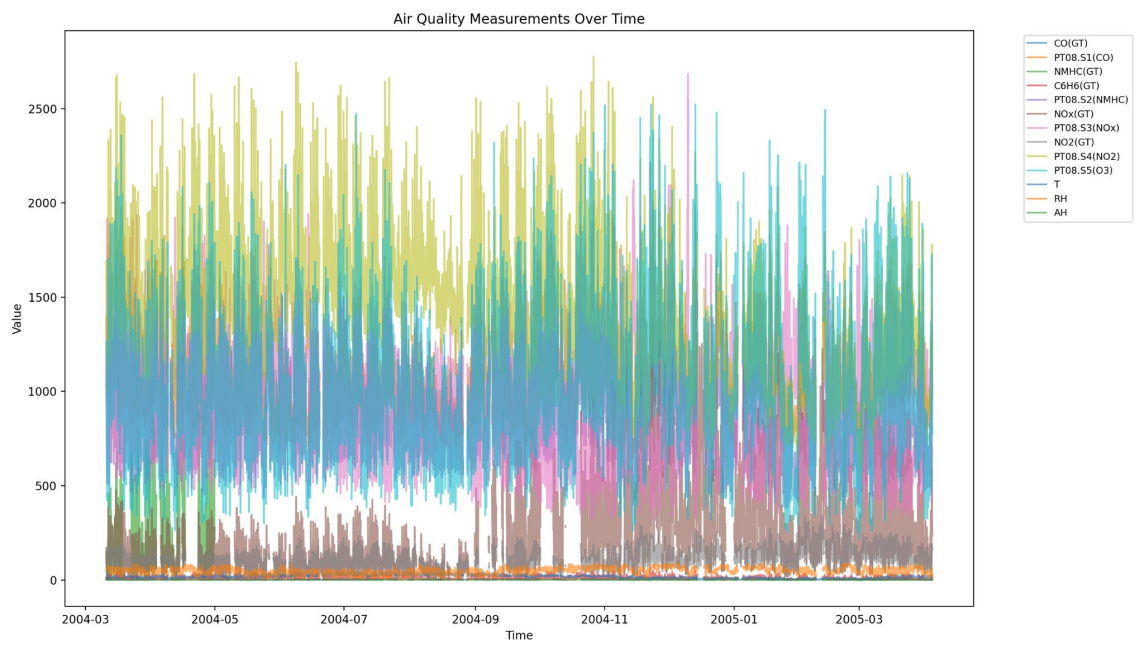


Perform **Linear Regression** to get correlation values and a predictive model



Perform **Pearson's correlation** to get correlation values and p-values to test statistical significance

Analysis Plan & Justification (2)



One of our EDA graphs. This one shows air quality measurements over the full year. Note the high volatility of the data.

Tricky Analysis Decision

Q: Which pollution measurements do we focus on?
A: All 13 of them!

There are a lot of different types of air pollution. At the beginning, we were wondering if we should focus on one or two types over the year, for simplicity's sake.

However, after doing EDA, we found that there seems a lot of variation for every pollution measurement. So we decided we would expand our project from "*do pollution levels correlate with day/night and weekday/end cycles?*" to include "*which pollution levels correlate most with these cycles?*"

This meant we had to do a little extra work, but we got to have a wider-breadth insight.

Bias & Uncertainty

Sampling Bias

All of the measurements came from hourly averaged responses from an array of 5 metal oxide chemical sensors embedded in an Air Quality Chemical Multisensor Device in a single Italian city, which may be biased due to the location.

Solution

Limit our conclusions to only being valid for that city. It was unnamed in the dataset, unfortunately, which makes it difficult to generalize our results. This also made it difficult to validate our models with current values.

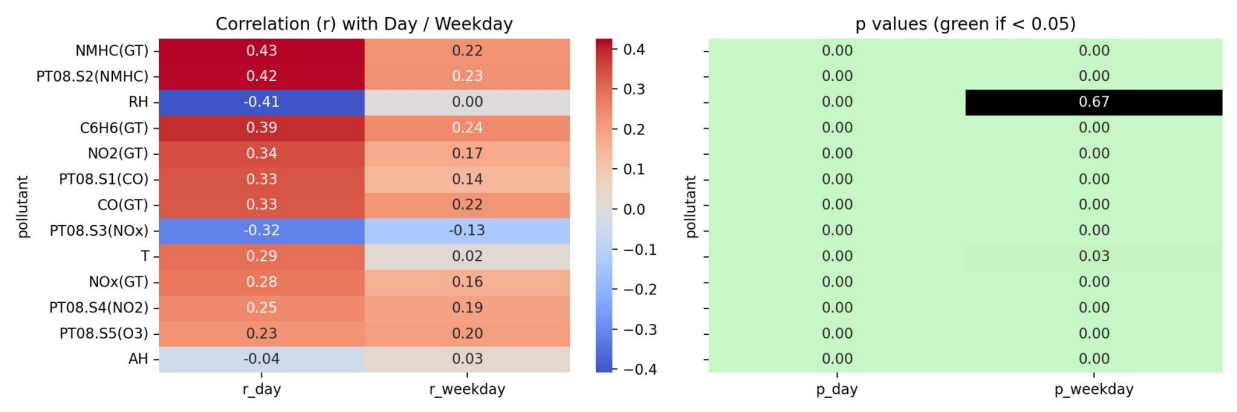
Results & Conclusions

Linear Regression

Test-split accuracy/eval metrics for each pollutant

target	n	rmse	r2	mae
PT08.S2(NMHC)	8991	235.082922	0.233425	187.168083
NMHC(GT)	914	164.607039	0.230368	127.138570
C6H6(GT)	8991	6.696062	0.219708	5.060480
RH	8991	15.903892	0.177731	13.322923
CO(GT)	7674	1.334764	0.144647	1.008114
NO2(GT)	7715	43.222138	0.144106	33.447549
PT08.S1(CO)	8991	200.939570	0.136780	158.873122
PT08.S3(NOx)	8991	234.300742	0.119383	180.825413
NOx(GT)	7718	192.840835	0.117712	145.922846
PT08.S5(O3)	8991	379.179789	0.102860	301.977511
PT08.S4(NO2)	8991	335.604973	0.098227	268.764814
T	8991	8.458658	0.086822	7.077280
AH	8991	0.405335	0.002403	0.331449

Pearson's Correlation



- Moderate to weak correlations across the board (0.43 to 0.00).
- NMHC(GT) and PT08.S2(NMHC) correlate the most with daytime
  - Both are measurements of Non Metanic Hydrocarbons.
- Day/night cycles correlate more with pollution levels than weekday/weekend cycles
- The predictive linear regression model is highly inaccurate on test data, probably due to high volatility of the data.
- Note: The linear regression correlation match Pearson's correlation values.

Next Steps

Test alternate hypotheses

Summer vs Winter?
Rush Hour?
Compare with other locations?

Collect a new batch of data

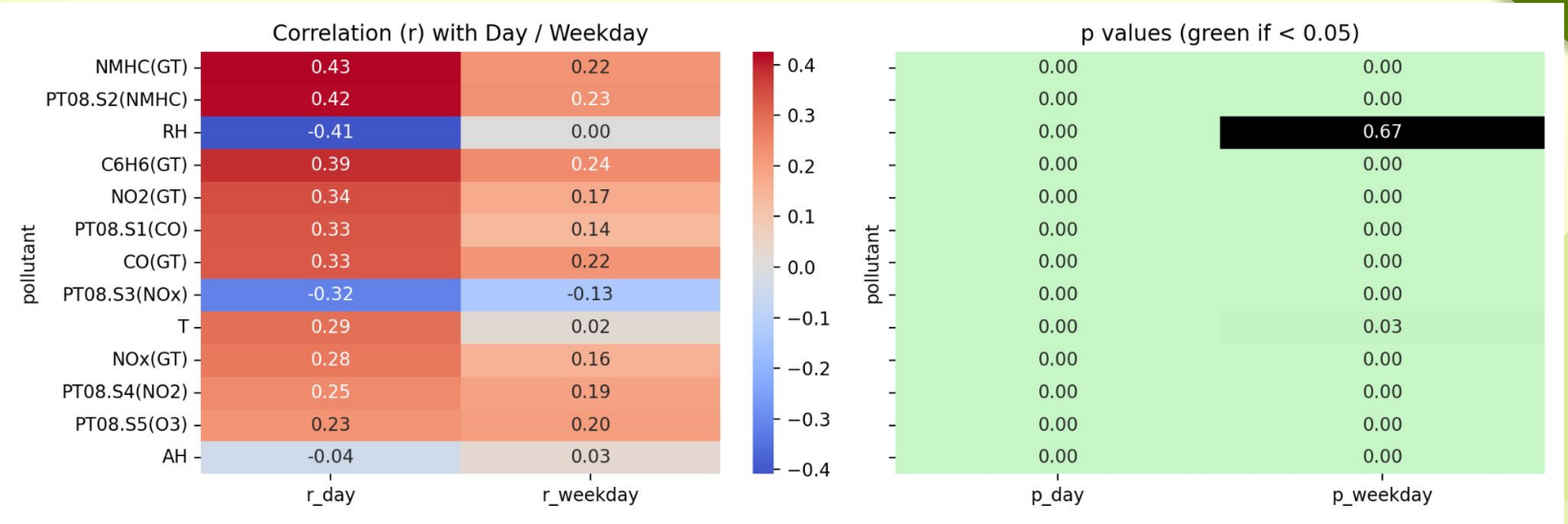
Redo the study in 2026 to compare patterns and hour by hour measurements.

References

[1] S. Vito, “UCI Machine Learning Repository,” *archive.ics.uci.edu*, Mar. 22, 2016. https://archive.ics.uci.edu/dataset/360/air+quality

Github: https://github.com/FlavienPMoise/ds4002-project2

Thank you!


Correlation (r) with Day / Weekday
pollutant
NMHC(GT) 0.43 0.22
PT08.S2(NMHC) 0.42 0.23
RH -0.41 0.00
C6H6(GT) 0.39 0.24
NO2(GT) 0.34 0.17
PT08.S1(CO) 0.33 0.14
CO(GT) 0.33 0.22
PT08.S3(NOx) -0.32 -0.13
T 0.29 0.02
NOx(GT) 0.28 0.16
PT08.S4(NO2) 0.25 0.19
PT08.S5(O3) 0.23 0.20
AH -0.04 0.03
r_day
r_weekday
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
p values (green if < 0.05)
pollutant
0.00 0.00
0.00 0.00
0.00 0.67
0.00 0.00
0.00 0.00
0.00 0.00
0.00 0.00
0.00 0.00
0.00 0.03
0.00 0.00
0.00 0.00
0.00 0.00
0.00 0.00
p_day
p_weekday